

Abhishek Pratap Singh

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Summary

Results-driven engineer with over **five years of experience** in **modem testing and integration**, specializing in the testing and Script developments with exposure in C and python programming languages, debugging and commercialization of different types of modem in 4G/5G products. Utilized expertise to collaborate with cross-functional teams & have overview on embedded Linux and Kernel internals.

Skills

Programming C, Python ,Embedded C ,Data structure

Technologies LTE, 5G NR (SA/NSA), NTN, NB-ntn, VoLTE, NB-iot, LTE-CATM1, IMS, GenAI

Tools & Protocols QXDM, APEX, R&S simulator/keysight, JIRA, VS Code, T32, Jenkins, Crashescope, TCP/IP,UDP, Modem protocols L1,L2 & L3, UART, GIT

Core Concepts 4G/5G-NR(Sub6 and mmWave), 3GPP specifications, Operating Systems, SIP, Memory Management, Knowledge of KPIs (Throughput, Latency, Handover, Reselection), System Programming

Soft Skills Modem validations & Integration , Debugging modem logs, Legacy Technologies, Script writing & modifications, Wireless Networks, Linux/Windows

Experience

Capgemini technology services india limited

Hyderabad, India

Software Engineer(Professional 1) | April 2021-present

- Being at client location **QUALCOMM** projected to perform functional and performance testing & integration of cellular IOT & android modem across 2G/3G/4G/5G ,NB-Iot and NTN technologies. Conducted thorough testing & log analysis with the cross-functional team, proactively monitor continuous integration testing and developed automation scripts for modem configuration and scenario execution using vendor -specific DSL.

Techveda

Internship & Training | September 2018- Oct 2019

During training period gained exposures onto Linux kernel internals ,Device drivers & embedded linux which emphasizes my skills on OSI model layers ,Board bring-up consisting through via BSP ,build root & Yocto initializations.

During Internship practically involved in internal projects using kernel modules ,hardware access, Interrupts ,multi-threading ,inter-process communication ,memory management ,signals management ,FILE I\O operations.

Projects

QUALCOMM

QTS110 Aware platforms product | Modem testing & integration, GSM, NB-Iot, CATM1

- Developed & executed automation scripts with concerned team for modem configurations and performed functional & performances validation of required sensors scenarios manually which includes specific functional teams collaborations. e.g MMODE, sensors, Core etc.
- Collaborated with OEMS and chipset teams for issues triage, modem stability ,log analysis and feature integrations.

QUALCOMM

SW5100 Atherton Wearable(snapdragon W5+ Gen 2) | NB-NTN, Modem validations & Integrations

- Conducted functional and regression testing for Wear OS-Based, ensuring compliance with Low-power states and battery life KPSs ,Led the integrations of different tech driven labels resultant validated power optimization ,connectivity features including LTE, WI-FI, GNSS ,GSM, WCDMA & VOLTE.
- Developed NB-NTN based automation scripts using DSL scripts to validated NTN specific bands & Performed backend mapping and integration functionality.

QUALCOMM

SW6100 ASPEN chipset (Snapdragon Wear W6) | 5G SA/NSA ,NTN , LTE

- Worked on Snapdragon Wear W6, a next-gen wearable SoC built from scratch as bring-up which consists modem stability reports to concerned tech teams ,logs analysis ,regressions report & follow-ups , functional issues debugging and reporting , maintaining power optimizations and configurations of devices performances.

- Implemented functional testing which includes handover, reselection, network registrations, mobility, data session establishment ,GNSS accuracy and throughput validations of modem.

QUALCOMM

SM4250-Snapdragon 460 and SM6225-Snapdragon 680 | DSDS, LTE, VoLTE, VoWiFi

- Started testing ,log analysis ,regressions debugging & automation scripts modification with required changes and as per need and add-ons too on this mid-range Qualcomm chipset that is optimized for gaming so that power optimizations & performances testing was the key areas to focus during testing. DSDS scenario with voice calls ,Voip , data calls & connectivity to validate end to end scenarios makes me to enhanced more into cellular domain.
- Implemented and modified the automation existing scripts as per the scenario testing requirement which includes content server & Iperf .

Education

Mahatma Jyotiba Phule Rohilkhand University, MJPRU

Bareilly,Uttar pradesh, India

B.Tech in Electronics & Instrumentation Engineering *2013 – 2017*

- Graduated with securing **6.5 YGPA**.

Govt. Jubilee inter college

Gorakhpur, Uttar Pradesh, India

XII – Science (UP state Board)

2012 – 2013

- Achieved **74.4%** in the Class XII board examination.

Certifications

Embedded Linux & Linux Kernel Internal systems | Tech Veda

GenAI chatGPT | Coursiv

Embedded C & Digital systems | ATI-EPI Hyderabad